

Artificial Intelligence in Deception: Unraveling the Role of AI and Large Language Models in Online Financial Scams and Countermeasures

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INTRODUCTION

Artificial Intelligence (AI) and financial scams provides both significant opportunities and challenges to the cybersecurity, consumer protection and ethical AI applications in the modernized world. In this study, I delves into on one of the most common scams - Phishing. Specifically, I focus on those generated solely by AI versus those crafted through a combination of AI and human customization.

Through a twofold methodological approach, I compare the patterns, the structure, linguistics and other characteristics in these types of phishing emails. Examining these types of phishing emails is to distinguish between the capabilities of AI algorithms and human creativity in crafting deceptive content.

By isolating the influence of AI in generating phishing emails, I can assess its ability to mimic human communication patterns and manipulate language for fraudulent purposes. On the other hand, studying partly AI-generated emails allows me to understand how human customization enhances the sophistication and effectiveness of phishing attempts beyond what AI alone can achieve.



METHODS

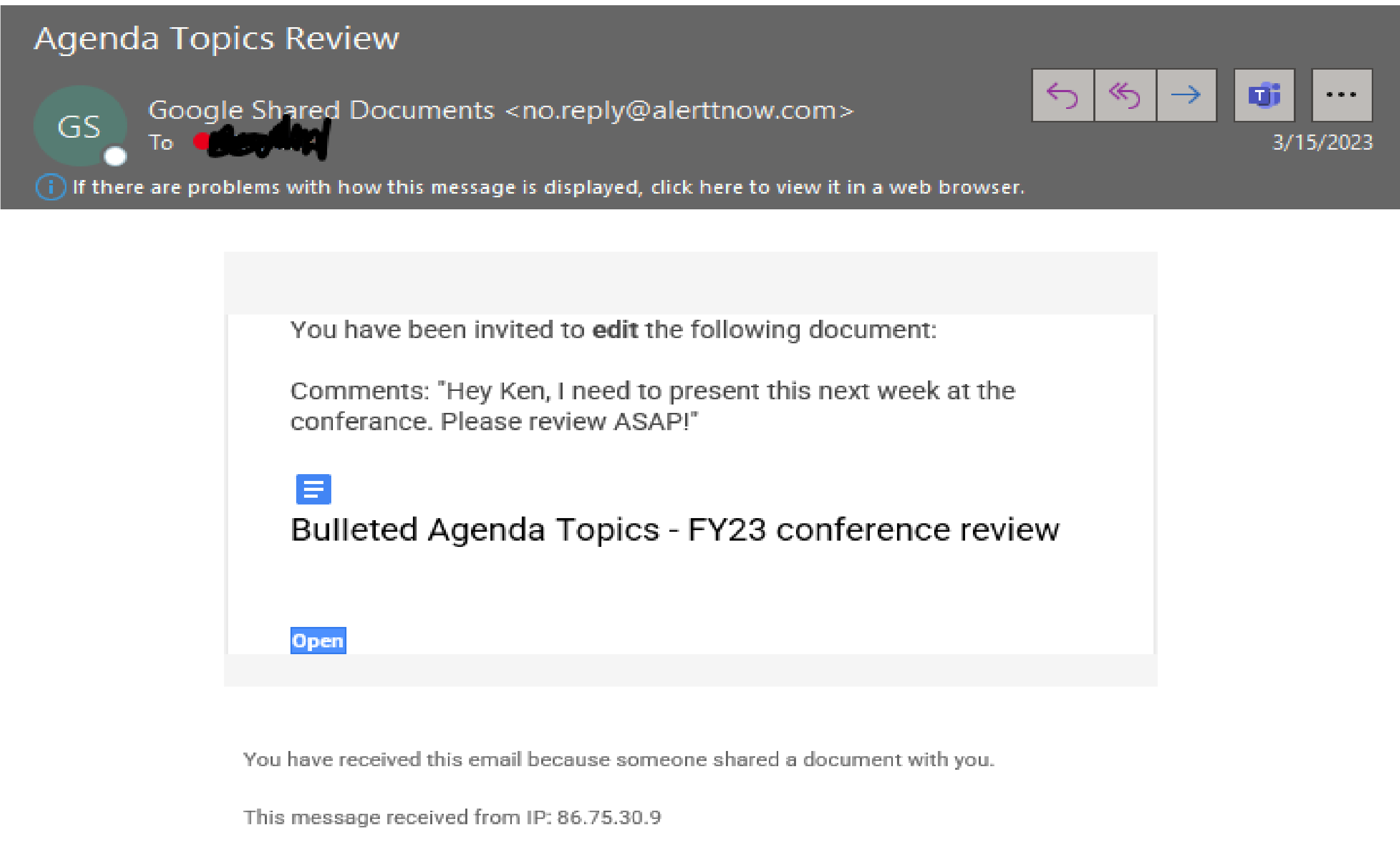
In this study, I used a combination of approaches to examine the impact of AI in financial frauds and the effectiveness of responses. My main technique was using chatbots specially ChatGPT and Gemini to generate phishing email simulations.

I created two phishing emails using Gemini, and three using ChatGPT, to offer a range of simulated phishing attempts. Additionally, I included data from UW-Stout Systems who utilized Proofpoint-created phishing emails that were later modified by humans. Phishing emails that were partially generated by AI can be compared to those that were generated fully by AI algorithms.

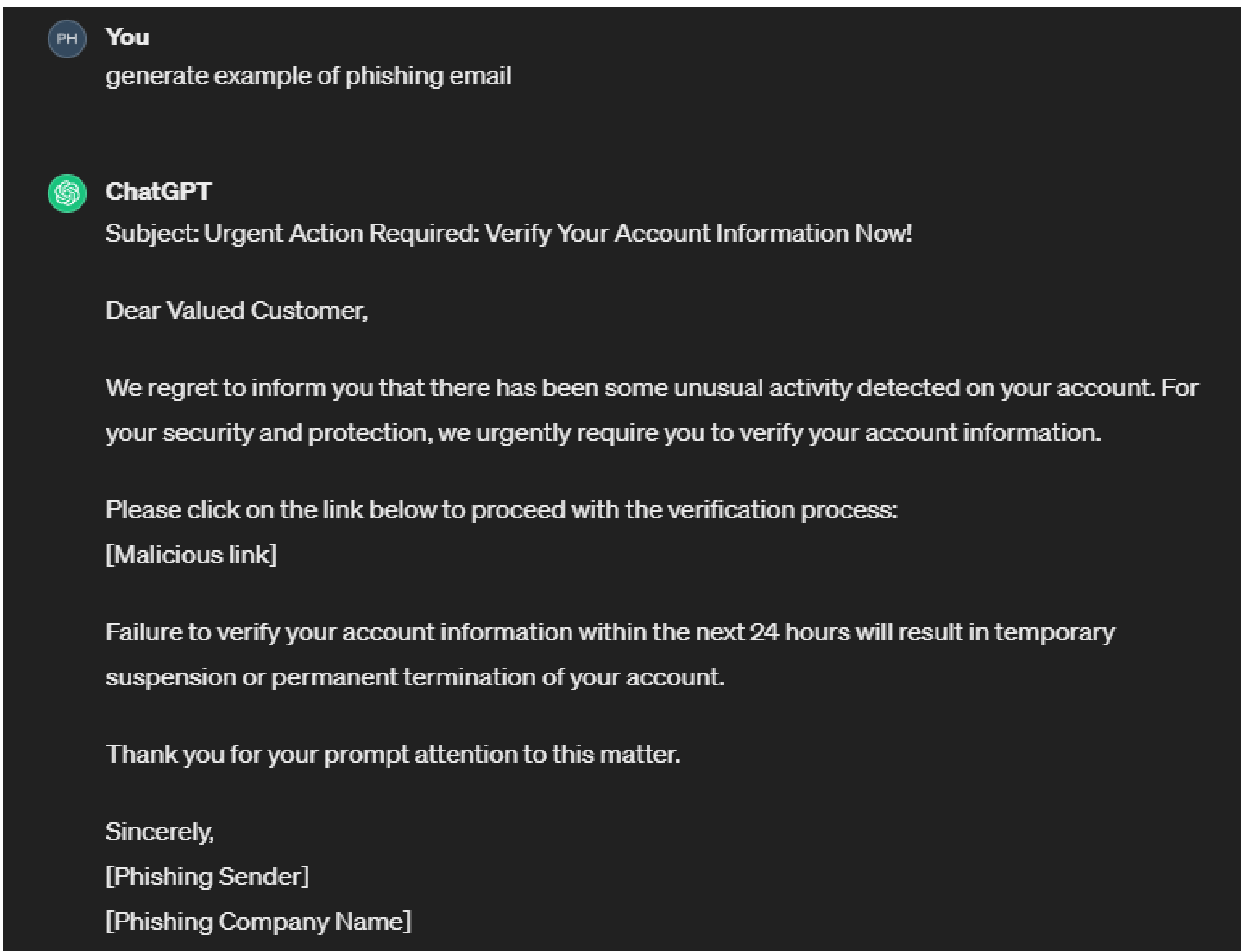
By analyzing the similarities, differences and effectiveness of phishing attempts, I seek to gain insights into how AI is being utilized to perpetrate financial scams through phishing emails to compromise data, personal information and identity to facilitate fraudulent transactions, unauthorized access to financial accounts and ultimately, to extract monetary gains and individuals and organizations.

EXAMPLES OF PHISHING ATTACKS

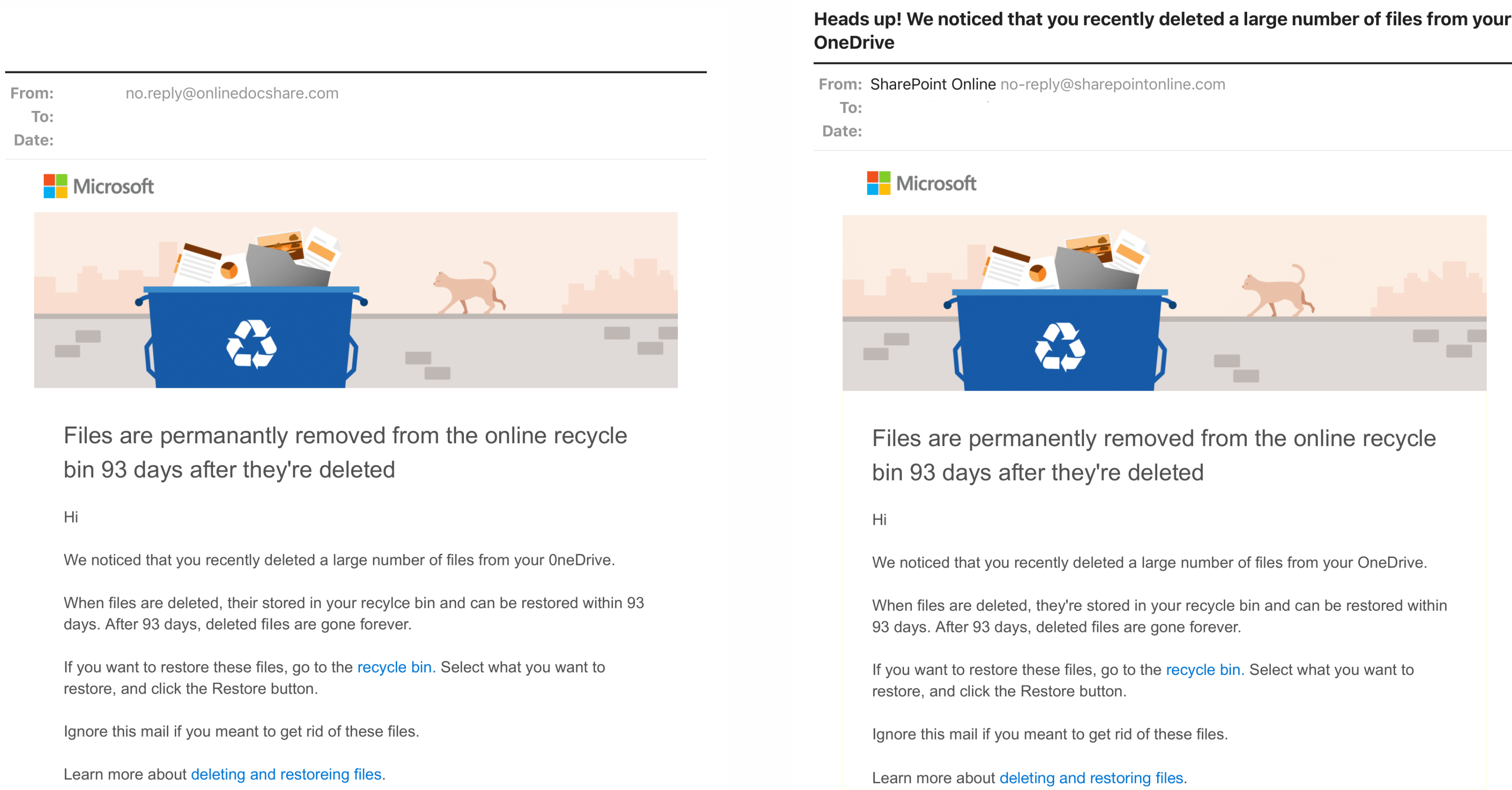
Partly AI



Solely AI



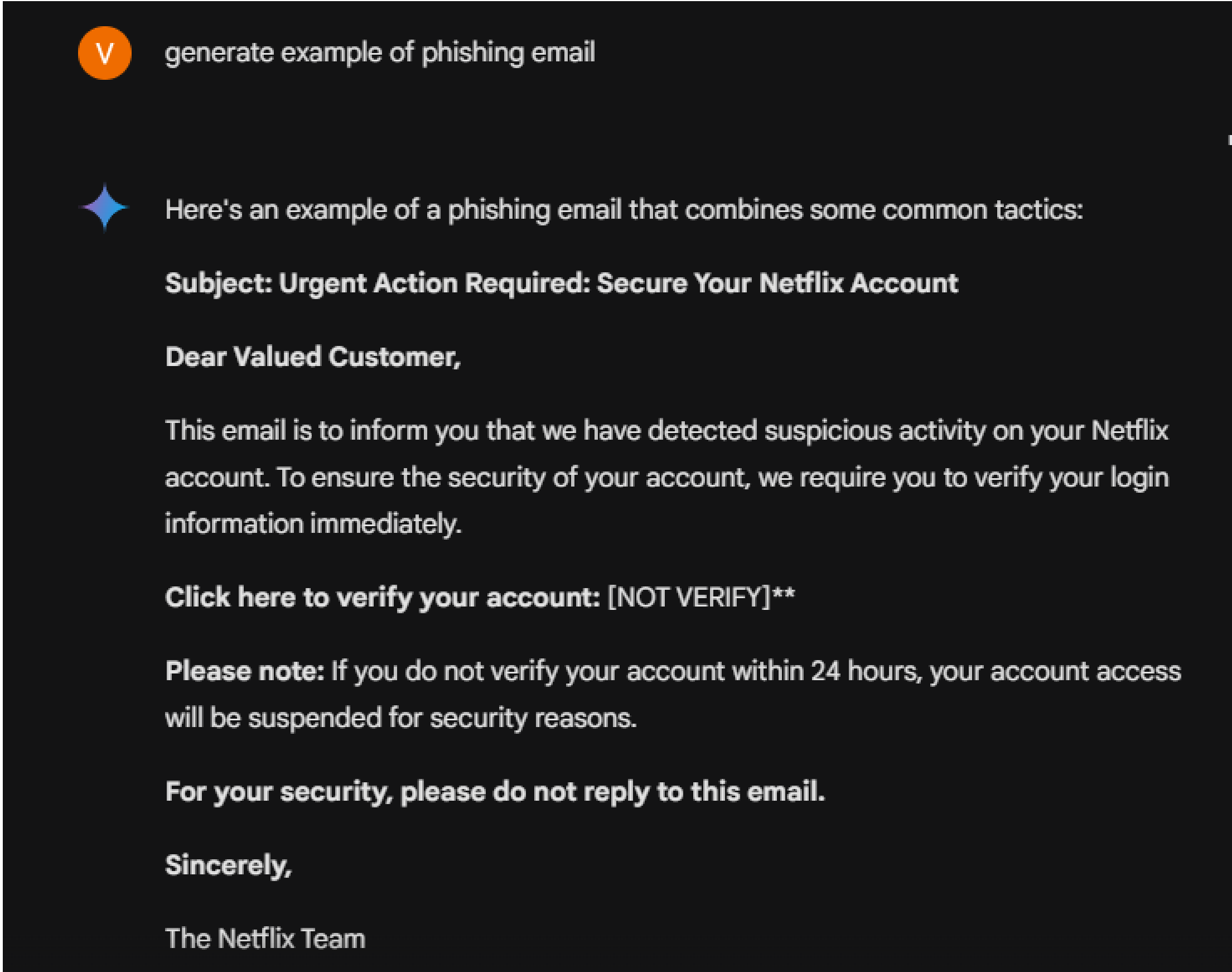
Example 1



Example 2

Bonus: Authentic Email Example

Example 1 – ChatGPT



Example 2 – Gemini

ANATOMY OF THE PHISHING ATTACK

Upon analyzing the phishing emails, it indicates that a significant majority adhere to a structured format comprising a subject line, recipient address, event's description, instructions, phishing link, urgency/threats, and sender information.

- Solely AI-generated emails exhibit a structured format with consistent elements, emphasizing urgency through subject lines and maintaining formal sender information.
- Solely AI-generated emails predominantly employ a direct approach, explicitly instructing recipients to "click on the link" within the email content.
- In contrast, partly AI-generated emails lack uniformity, showing variability in structure, sender profiles, and linguistic conventions.
- Partly AI-generated emails often lack urgency indicators in subject lines, contain errors like spelling mistakes, and may resemble spam rather than targeted phishing attempts.
- The absence of formal salutations and closing signatures further distinguishes partly AI-generated emails from their solely AI-generated counterparts.

These findings underscore the challenges of detecting and mitigating phishing attempts, particularly as human interference introduces variability in email composition. This variability impacts the structure, language, and sender profiles of phishing emails, resulting in partly AI-generated emails exhibiting less predictability and posing increased difficulty in detection.

REFERENCES

ChatGPT 3.5 version, OpenAI, 10 Mar. 2024, chat.openai.com/chat.
Gemini, 10 Mar. 2024, gemini.google.com
"Phishing Campaigns". University of Wisconsin KnowledgeBase. University of Wisconsin Technology Help Desk. Access Date: 8 Mar. 2024

